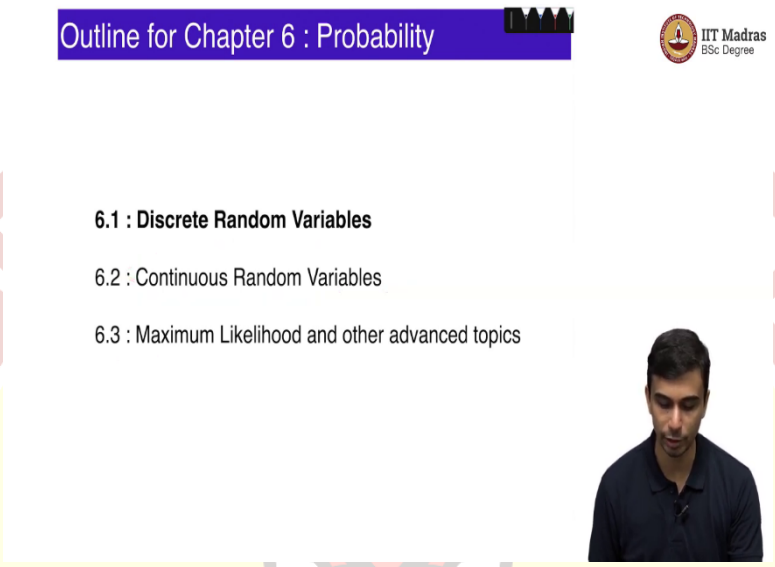


IIT Madras

ONLINE DEGREE

Conditioning
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Chapter 6: Probability

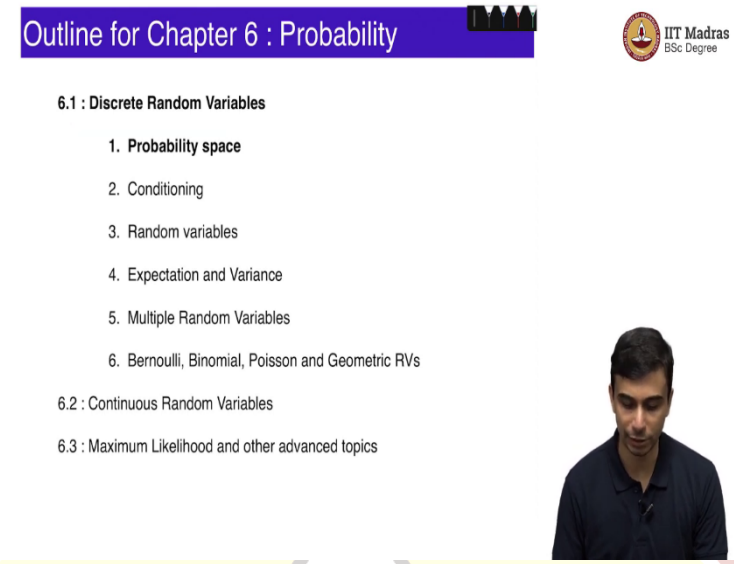
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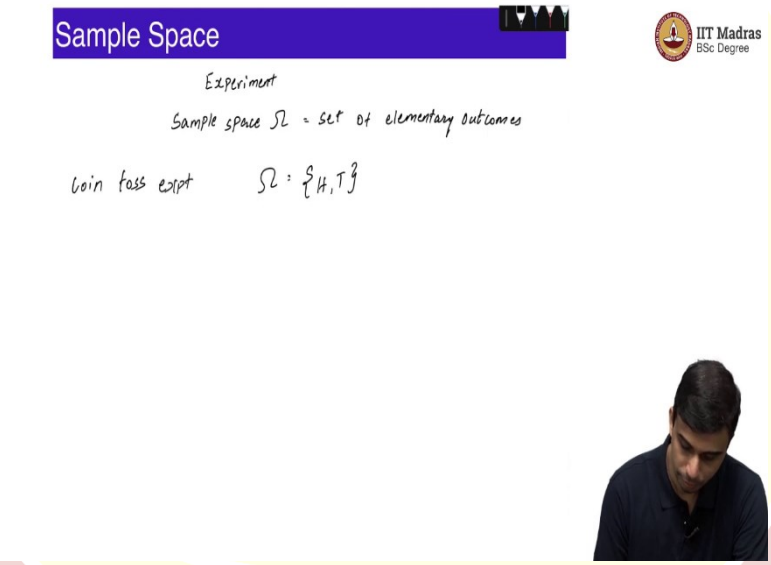


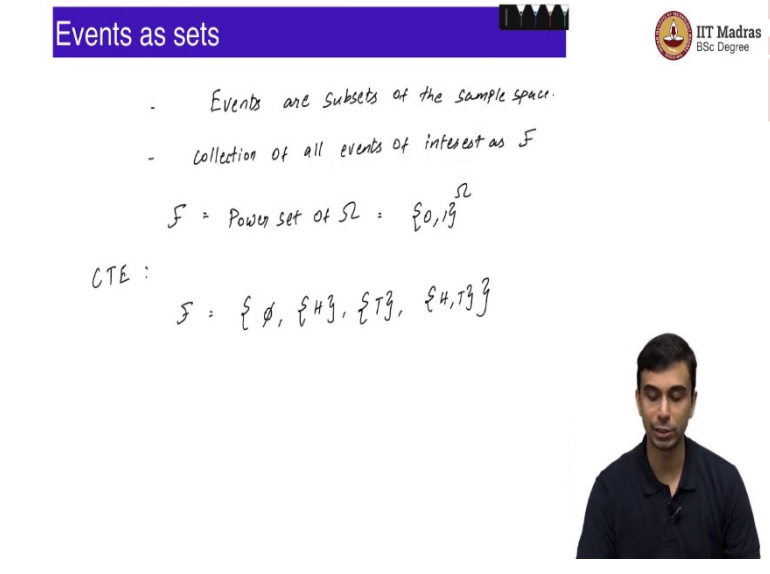
Hello, everyone, and welcome to another chapter of the Machine Learning Foundations course. In this final chapter, we will see the basic probability tools that will be used throughout machine learning. And towards the end of this chapter, we will see, one of the examples of unsupervised learning that we covered in the first week, which was density estimation. And also touch upon another related unsupervised learning problem of clustering.

This chapter, we have divided into 3 modules. The first module will just cover the basic, simple discrete random variables and move onto continuous random variables, and wrap up with advanced topics like maximum likelihood and Gaussian mixture models.

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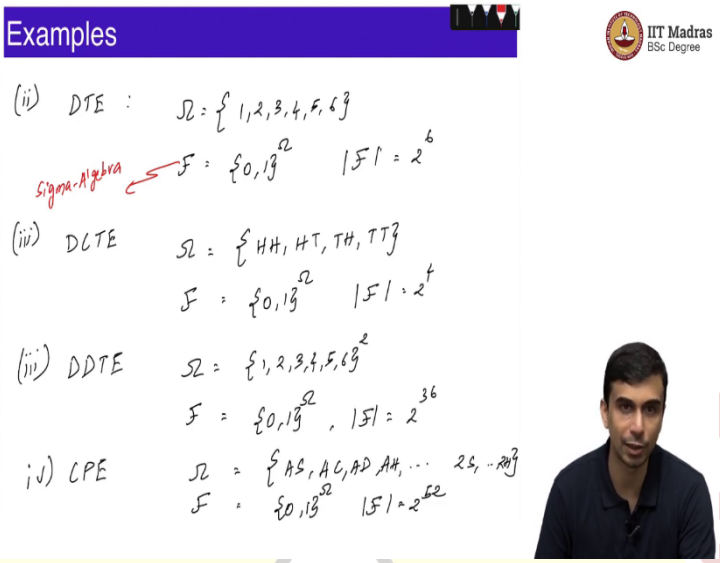
In the first week, we will just take a look at simple discrete random variables. Most of this might be familiar to you from either stats for data science course one or two, but we will go over this again. The first basic tool or the basic building block of probability is the experiment. All the statements in probability are in the context of what is called an experiment. And every experiment has a set of basic elementary outcomes, which are all collected into a sample space. Sample space, which is typically denoted as ω is the set of elementary outcomes.

For example, consider the experiment of tossing a single coin in such for such an experiment, the two basic outcomes are head and tail. So, in which case ω would simply be H comma T, for the coin toss experiment. The sample space is the set of all elementary outcomes, and that we are able to express all quantities of all events or events of interest as essentially subsets of the sample space. Events are subsets of the sample space.

We will denote the collection of all events of interest as a F, collection of all events of interest as script F. In the case of discrete random variables, where our ω is simply a finite set, we will simply take F to be equal to the power set of ω, which is denoted as 0 comma 1 power ω, that is, the cardinality of the set F now is simply 2 power the cardinality of ω.

For example, in the coin toss experiment, we have F would simply be equal to the null set, the singleton H, the singleton T, and the entire sampled space. For the case of finite sample space, which is going to be the focus for the first week. We can take without any issue the set of all subsets of ω to be the set of all events of interest. So, the events of interest is not particularly important right now, if you do not understand, if you do not know why we are using events of interest, it does not matter as of now. As of now, the sub sets of ω forms are events.

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So, here are a couple of examples. We saw the first example that was the coin toss experiment. Let us have a second example is dice throw experiment. Let us call that DTE. The dice throw experiment we can have ω to be six possibilities. It is either 1, 2, 3, 4, 5 and 6 and the set of events of interest F is simply the power set of ω which is 0 comma 1, power ω or have another, consider another example, as the double coin toss experiment.

You toss two coins, so let us call it double coin toss experiment in which case, you have the result of two coin tosses, in which case the sample space we have four possibilities, which is either HH, HT, TH, TT. Once again, we have our set of events of interest as simply 0 comma 1 power ω. In the first DTE experiment, our cardinality of F, the set of all collection of all events is simply 2 power 6. In this case, it is going to be 2 power 4.

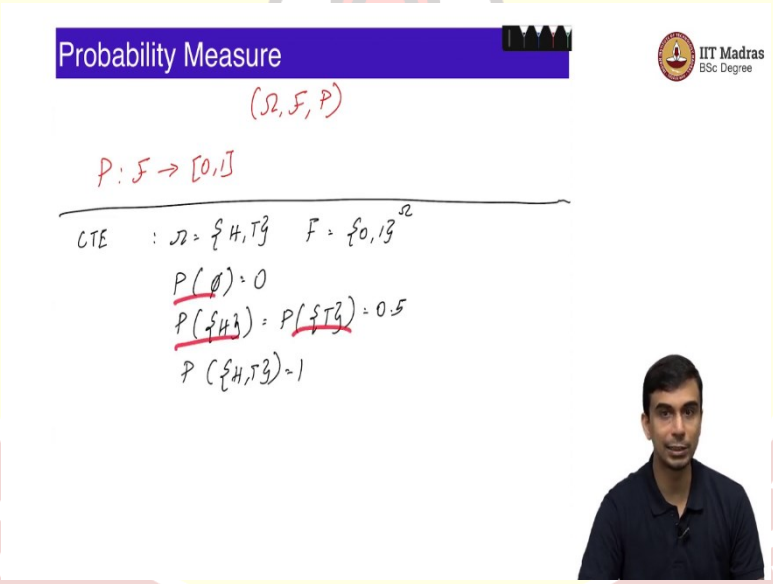
We can do the same thing for the dice, as well, we can have double dice through experiment in which case ω is 1, 2, 3, 4, 5, 6. The whole square I mean, it is a cartesian product of 1 comma I mean elements of the sample space here would look like 1 comma 1, 1 comma 7, 1 comma 5, 4, comma 6, and so on, so there are 36 possible elements in the sample space. Once again, we are going to be using our set of events of interest as simply 0 comma 1 power ω.

And in this case, the cardinality of all the events of interest would be 2 power 36. So, these are some of simple. Let us go further and consider one more experiment. Let us call that the card picking experiment. You are picking one card at random from full suite of 52 cards. So, our ω would be something like let us say the ace of spades, the ace of clubs, the ace of diamonds, the ace of hearts, and all the way to, let us say, two of spade, two of hearts and so on. So, you would have 52 would be the cardinality of the sample space.

Once again, the simple F would simply be 2 power ω 0 comma 1 power ω, the cardinality of which is going to be 2 power 52. The cardinality of 2 power 52. So, these are some simple examples that you would have seen in any probability or statistics course in your school or the earlier courses. But we will frame them in the context of sample spaces and the set of relevant or events of interest.

In some cases, we will call this set F, we call the sigma algebra. So, for now, they are not of interest for us because we are simply using the power set of the sample space as the set of events, but in more complicated cases where you cannot do that. For example, if your sample space was in finite, then the power set of an infinite set is tricky to handle. So, we will not do that actually. So, we will do something else. But so, then it must satisfy certain properties, then the sigma algebra must satisfy certain properties, but they are not of interest right now to us because we are using the power set as the planning example.

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Now, we will move on to the final concept, final constituent of the experiment. We have already, the experiment is defined by three things, which is ω, which is the sample space, the sigma algebra F and the third element is what is called a probability measure. A probability measure is basically a function from the set of events, the collection of events to a positive number. In fact, you do not need to say positive number it is between 0 to 1.

A probability measure is simply something which takes in an event and outputs a number between 0 to 1. You can eventually, you can effectively view this number, as the probability of that event happening. Every element in F is simply can be viewed as an event. And the

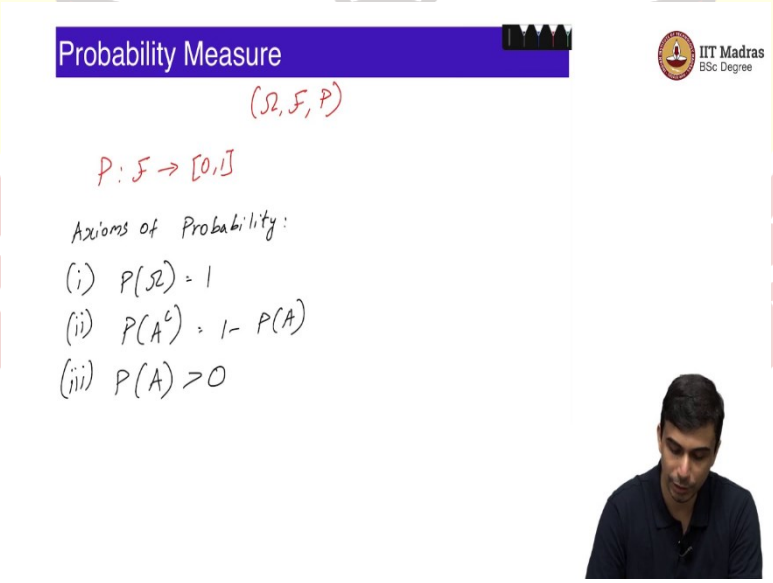
probability of that event is given by the probability measure. We will once again go over the examples and give the probability measure for these first.

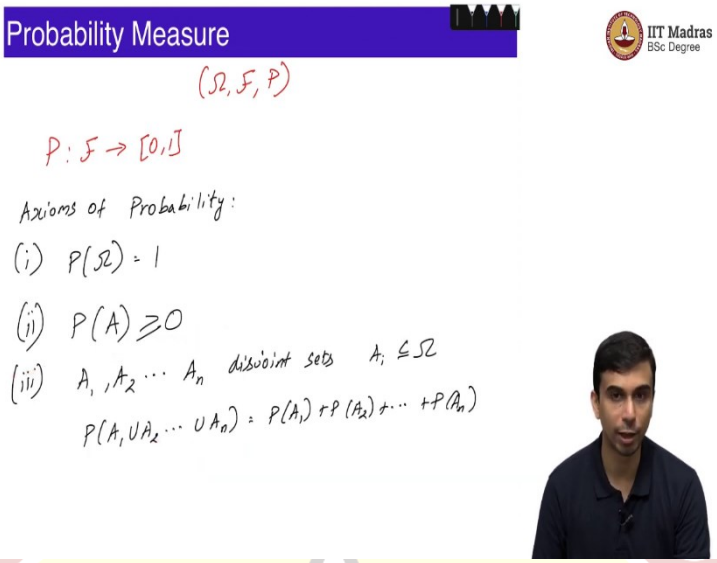
So, let us take the first example that we had, which was the coin toss experiment. We had ω was H comma T and F was simply 2 power ω or 0 comma 1 power ω. We can have multiple probability measures, valid probability measures, but depend on the situation and reality at hand.

For example, if this could be a third coin, in which case, probability of the null set would be 0, probability of H equal to probability of T would be equal to 0.5. And probability of H comma will be equal to one. So, this because the sigma algebra F only contains four elements, just by defining the four values here you completely define the probability measure. And the probability measure is simply a function which takes in an element from the Sigma algebra F, and outputs a valid number. So, this is an example probability measure.

In general, it could be any function, but it must satisfy certain properties. What are the properties that the probability measure must satisfy? So, these are called the axioms of probabilities are very intuitive properties that the probability measure should satisfy. Let us write that down.

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Let us write that down first. The properties of the axioms of probability are as follows. First axiom is the probability of the sample space is equal to 1. We call that ω is always going to be in the sigma algebra F, simply because it is the power set as far as concerned, we are concerned right now, F is a power set, and the entire set is also an element of the power set. The entire sample space, so probability ω should be equal to 1, that is the first property.

And the second property is that probability of A complement should be equal to 1 - probability of A. Well, this is not strictly a property of its own independent it is a property corollary. The real property, is that, the probability of any set A should be greater than 0. Let us change this property to property 2. Once again, it makes sense that the probability of any event should not be negative, so it should not be greater than or equal to 0, it can be 0. So, the property of any event cannot be negative, so that is be the second problem.

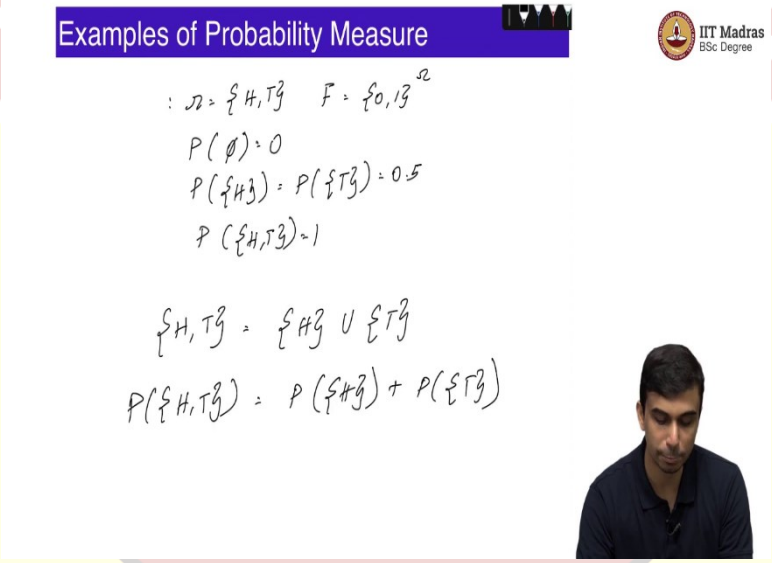
And the third property which is perhaps makes the definition makes obvious sense as a probability function, which is that if you have A1, A2 dot, dot, dot An disjoint sets and all of these sets are in F are Ai, all of these are subsets of the sample space. And by definition, these are going to be inside the sigma algebra, these are going to be element of the sigma algebra.

If A1 to An are disjoint sets, then you have probability of A1 union A2, dot, dot, dot, union An should be equal to probability of A1 + probability of A2 + a dot, dot, dot probability on An. So, the probability of the union of disjoint sets, this should be equal to the probability of a sum of the probability of the individual sets. So, this is a key property of the probability function or the probability measure.

And this particular statement can also be extended to infinitely countable sets also. But for now, we will just stick to finite sets because we are going to have our sigma algebra F itself is going to be finite, because our ω is going to be finite, which means that the sigma algebra is also going to be finite. As far as we are concerned, right now.

When we move on to continuous random variables and more complex settings, this is not going to be true. But for now, it is going to be, it is a simple and the probability of union of disjoint sets should be equal to the sum of the probability of the sets. So, that is the third and final axiom of probability.

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So, let us analyze the first probability measure that we had here. Is this a valid probability measure? Well, we do have that the probabilities are all, none of the problems are negative, so the second axiom is satisfied. And then the probability of the entire sample space is exactly equal to 1 that is also satisfied. And we have probability of null set should be equal to, that is, that should take any two sets. Take let us say you can write the set H, as H union null, and null is clearly does not, null is not a subset of anything.

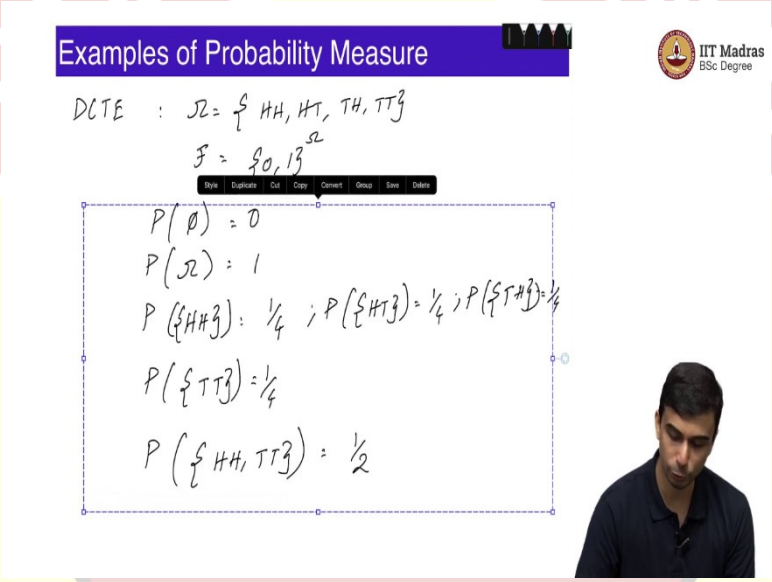
So, sorry, knowledge at this time. So anyway, so probability of H should be equal to probability of H + the profit null set. So, that means, that the property of null set should be 0, so that automatically it is satisfied here. And similarly, if you take the set HT, you can write the set HT, as a disjoint union of the set H and the set T.

So, left hand side, if you take the property to the left-hand side, that is probably a HT that is equal to 1, and is equal to the probability of H + probability of T which is 0.5 each. So, this

particular probability measure does indeed satisfy the axioms of probability. So, probability of H comma T the set H comma T is indeed equal to the probability of the set H + the probability of set T.

So, here is a simple exercise for you. Can you come up with a probability function, which does not satisfy this property? An easy way it would be to just simply change this. So, if instead of 1 make it 0.8, this clearly violates the axioms of probability. So, that is a simple exercise for you, I mean, we can think of other ways in which this probably axioms can be violated.

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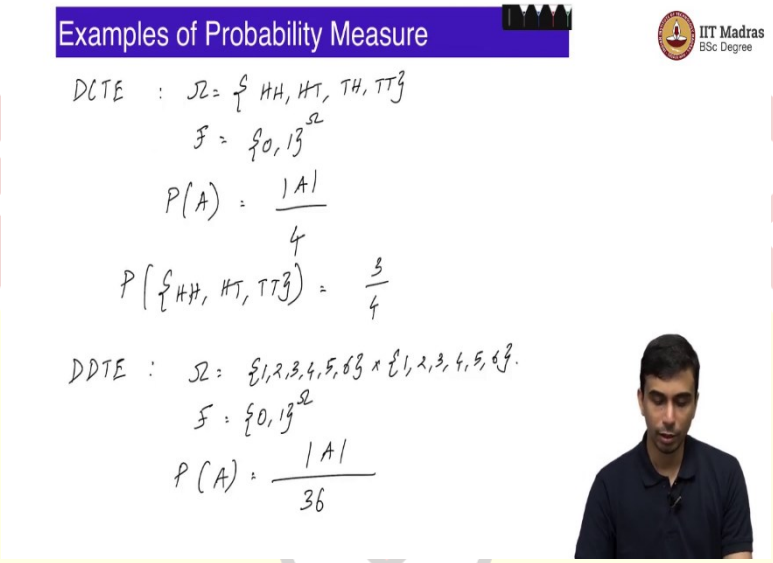
So, one more, we will give one more example. Let us take the double coin throw experiment in which case, the sample space is consists of four elements HH, HT, TH and TT. Now, our sigma algebra F is the power set which consists of 2 power 4 elements. Let us actually define all possible things first.

Probability of null is equal to 0, is an example probability measure. So, probability of ω should be equal to 1. What is a candidate example of probability of the set HH? Let us make it 1 by 4. Probability of the set HT is equals to say 1 by 4, and probability of the set TH is also equal to 1 by 4, and probably have TT is also a good 1 by 4.

In fact, by just defining these four probabilities, well, let us finish the fourth thing also. We have actually defined the probability for the rest of the elements of F also. Recall, F contains 2 power 4, which is 16 elements. I have defined given probability values only for, well, these two are trivial anyway, we have only given probability values for four values, but by the law by the axioms of probability I have also given the probability for the other things.

What is that? What is the probability? You can use the axioms of probability to answer this question. What is the probability of the set HH comma TH, TT? What is that? Well HH, the set HH comma TT is the union of the singleton HH, and the union singleton TT, which means that by the axioms of probability, it should be equal to 1 by 2, 1 by 4 + 1 by 4 it equal to 1 by 8. So, let us generalize this and write the probability measure in a compact way as follows.

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We will consider this particular probability measure that the probability of an event A is simply equal to the cardinality of A by 4. Why is this a valid property measure? Immediately, you can see that probability of a null set is going to be equal to 0 and probability of ω is equal to 1, so it satisfies the probability of ω is cardinality of ω, which is 4, 4 by 4 is 1, so it immediately satisfies those two axioms.

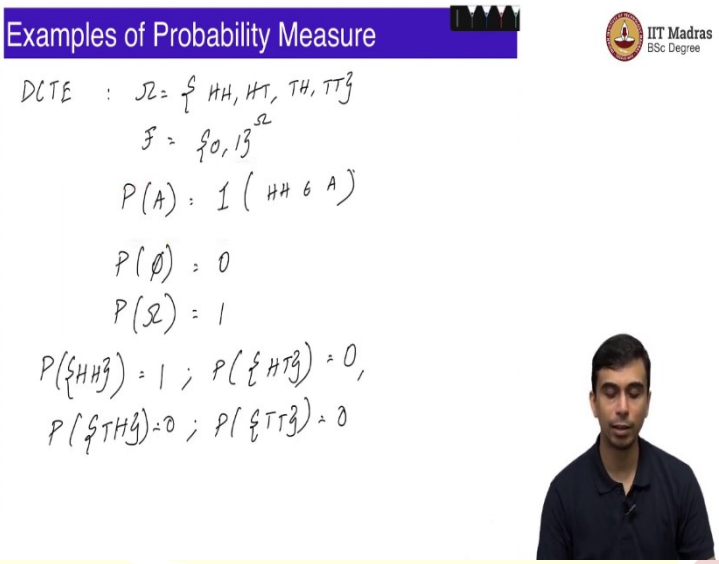
And it also satisfies probability of singleton HH to be equal to 1 by 4, singleton HT to be equal to 1 by 4 singleton TH equal to 1 by 4, probability of the set HH comma HT, comma TT. Well, this is also a valid event. And the cardinality of the event is simply 3 and 3 by 4. So, this is the probability to measure according to this particular probability measure.

Is this the only valid property measure? No, it is not. There are other valid property measures also, but this is the natural probability measure for a lot of situations. So, this is something that you should keep in mind. Let us have one more example of a probability measure. Let us say a double dice throw experiment, where you have your ω is 1, 2, 3, 4, 5, 6 the whole square. So, they are cross 1 comma 2 comma 3 comma 4 comma 5 comma 6. So, this is your sample space, and F is simply the power set of ω 0 comma 1 power ω.

Once again, let us consider the simple probability measure given by probability of a set A is simply the cardinality of the event A divided by 36. Is this a, first of all, is this a valid probability measure? Well, it seems so. First of all, you can substitute A equal to null set, which would mean that probability of null set equal to 0, and probability of the entire sample space ω is equal to 1.

And in fact, you can just substitute any set, any event to the subject of ω, and you can get a valid property measure, and it does satisfy the axioms of probability, again, it is pretty easy to check that the satisfies axiom of probability. So far there are other. Once again, we will see other probability measures for the same sample space. So, let us do that now. Let us actually do that so that. Let us take the double coin throw experiment.

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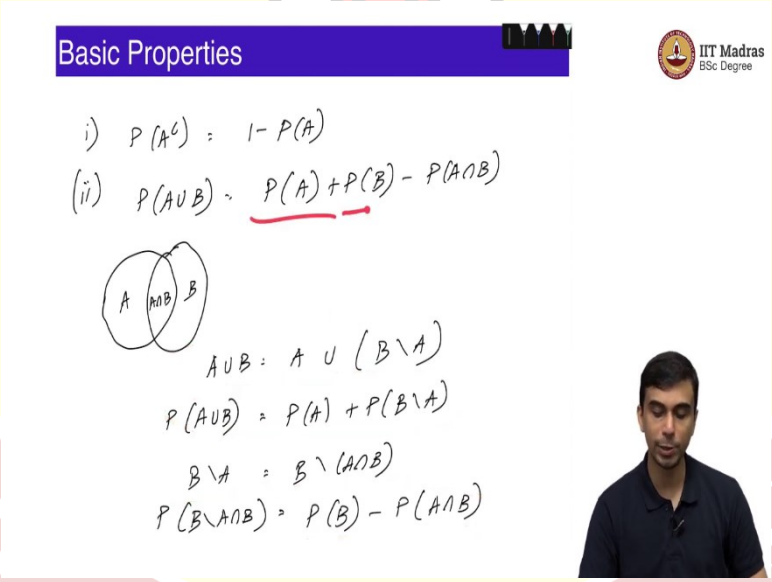
Let us take the double coin throw experiment, and give another probability measure, which is also valid. And we will say probability of A is equal to this is the probability, which I am going to use, that is, the probability of any event A is equal to 1 if HH belongs to A, it is going to be 0 otherwise, that is what the indicator random variable here gives, which is equal to 1 or 0. So, this is the indicator value indicator function.

So, is this a valid probability measure? Once again, again, how probability of null set is equal to 0? Probability of the ω is equal to 1 because HH does belong to the sample space. That is fine. And what about the elementary sets? Probability of HH, the set HH is equal to 1 because HH belongs to the singleton HH. What about the probability of the event given by a singleton HT? What is that? Well, that is going to be 0 because HH does not belong to the singleton HT.

Similarly, for every other singleton probability of TH is also the singleton TH is also equal to 0, probability of the singleton TT, is also equal to 0. What is if you can use these four elementary probabilities to get the probability of any set? For example, what is the probability of the set HH comma HT? Well, that is equal to 1 because HH, the probability of H is equal to 1, and probably the HT equal to 0, so you can sum the probability of the disjoint events, which is 1.

What does this experiment really correspond to? This experiment really corresponds to the case where the coin happens to be a Sholay coin. So, it will always give heads only. So, that is what this particular experiment corresponds to. So, depending on the probability measure the experiment can be, even though it is a normal coin, and the Sholay coin are both coin toss experiments the probability measure tells whether it is a fair coin, which corresponds to this, or whether it is a Sholay coin, which corresponds to this.

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So, that is examples of property measure. So, what properties should the probability measure satisfy? So, we have already seen the basic axioms, and those axioms imply certain other properties. Those are pretty easy to see, which is the probability of the set A complement is equal to 1 - probability of A. Why is that? Because A complement and A, A union A compliment is equal to the entire sample space ω and A and A compliment are disjoint.

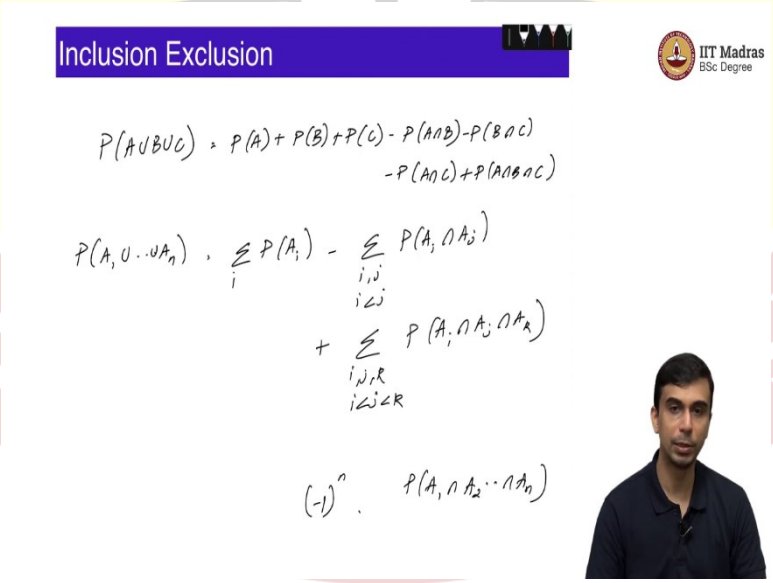
So, probability of A + probability of A compliment is equal to probability of ω, which means, that probability of A complement is 1 - probability of A, that is also a meaningful, it is a ECD interpretable statement here. We can extend this even further and say probability of A union B.

What is probability of A union B? Well, probability of A union B is equal to probability of A + probability of B - probability of A intersection B. Why is this true? Well, figuratively is this easy to see, you can see with a figure, this is A this is B this is A intersection B. So, if you can view it, probably A union B as A union, B difference A. B difference A and A are disjoint sets, so you can write this as probability of A + probability of B difference A.

Now, we can write probability of B difference A. Well, first change the set. What is B difference A? B difference A is really the same as B difference A intersection B or as far as B difference A is concerned B difference A is exactly the same as B difference A intersection B. The advantage with difference A intersection B is that probability of B difference A intersection B is exactly equal to probability of B - probability of A intersection B.

Why is that? Well, that is because B difference A intersection B union A intersection B is exactly equal to B, and these two are disjoint sets. So, probably a B difference A intersection B is exactly equal the probability of B - probability of A intersection B, and you can put both of all this, this equation and this equation together to get probability of A union B as simply equal the property of A + property of B - probability of A intersection B.

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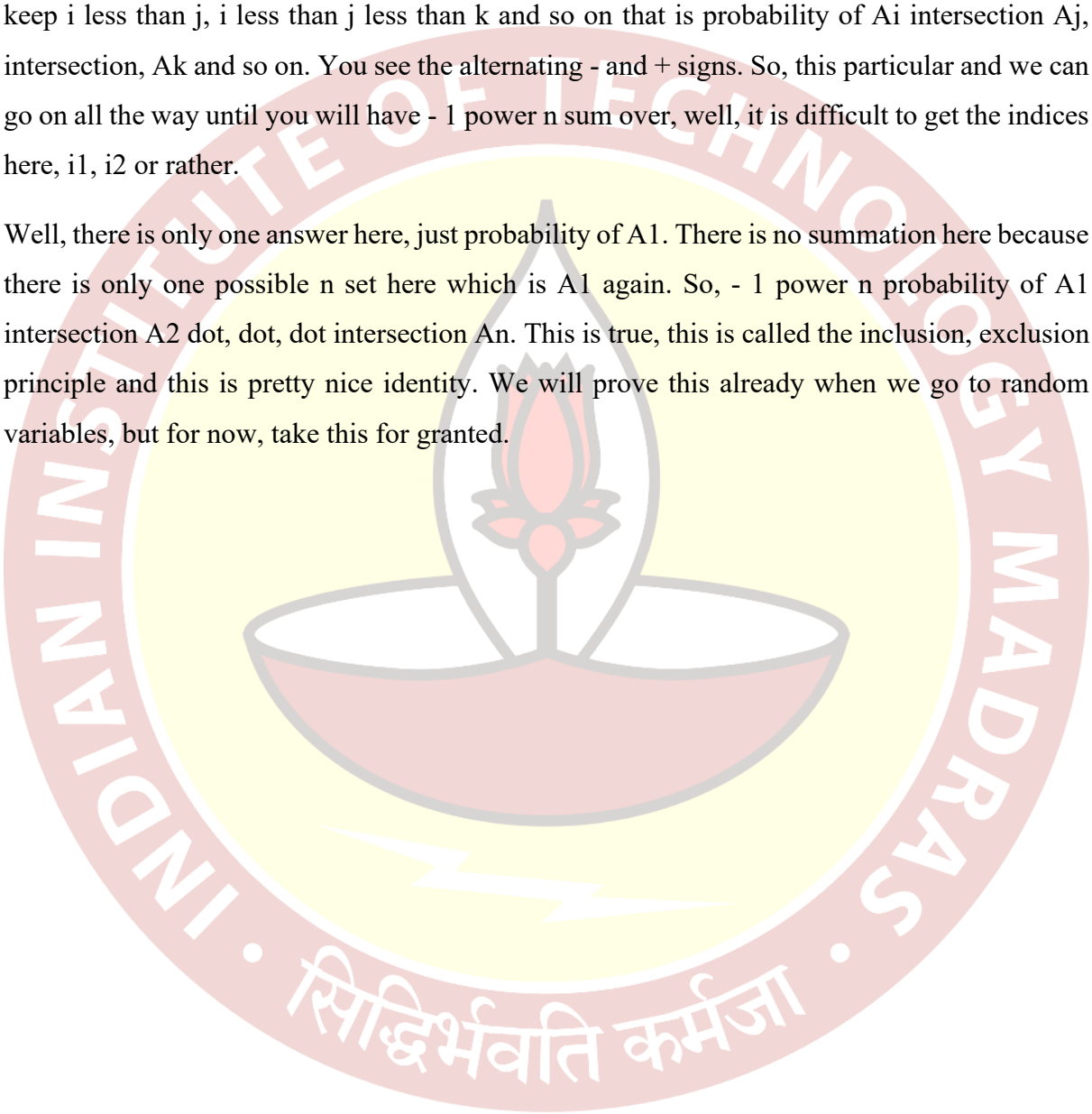


In fact, this can be extended even further for multiple unions. For example, you can say probability of A union B union C is equal to probability of A, + probability of B + probability of C - probability of A intersection B - probability of B intersection C - probability of A intersection C, + probability of A intersection B intersection C. So, this seems more complicated, but you can see the pattern here. You are adding the single elements and then subtracting the double intersections and then adding the triple intersections and so on.

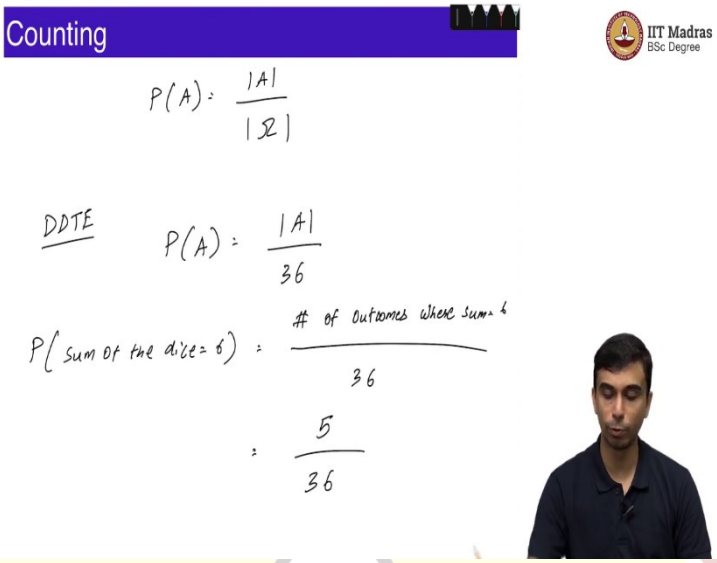
So, in general, if you have probability of A1 union dot, dot, dot An, you have n sets, this is simply equal to sum over, i probability of Ai, so this is the singleton probabilities you are adding it up, and then you subtract sum over i comma j, i less than j, let us say probability of Ai, intersection ij, and then you add sum over i comma j comma k, i less than j less than k.

So, this i less than j less than k so that you do not double count intersections, and so you always keep i less than j, i less than j less than k and so on that is probability of Ai intersection Aj, intersection, Ak and so on. You see the alternating - and + signs. So, this particular and we can go on all the way until you will have - 1 power n sum over, well, it is difficult to get the indices here, i1, i2 or rather.

Well, there is only one answer here, just probability of A1. There is no summation here because there is only one possible n set here which is A1 again. So, - 1 power n probability of A1 intersection A2 dot, dot, dot intersection An. This is true, this is called the inclusion, exclusion principle and this is pretty nice identity. We will prove this already when we go to random variables, but for now, take this for granted.



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So, there is one particular probability measure, which is, which you will see very often. We have already seen it twice, which is that, you have the probability of any event A is simply equal to the cardinality of A divided by a cardinality of the sample space. This particular probability measure is effectively called the equal probability. Every event has, every elementary event in the sample space has the same probability, that is equal to 1 by the cardinality of the sample space.

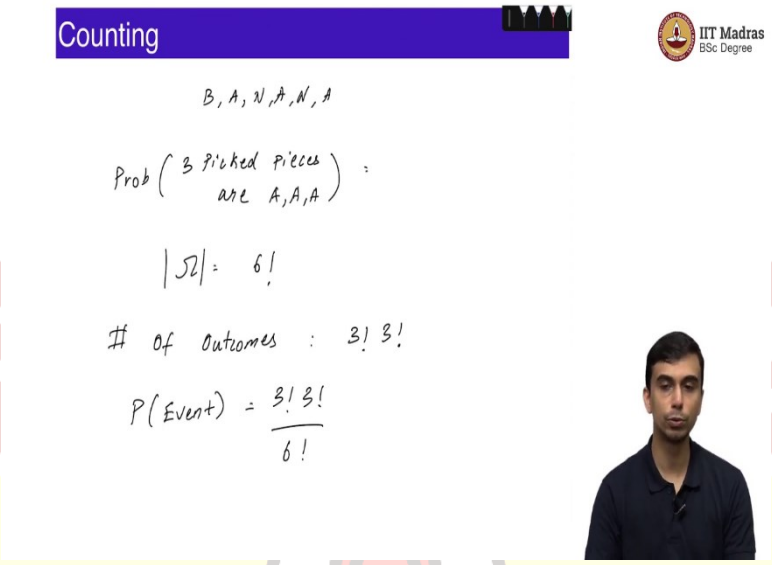
In this particular probability measure, essentially measuring probability is exactly equal to counting. Counting how many, counting the probability of any event is simply counting. Measuring the probability of any event is simply counting the number of times that event happens. For example, let us take a simple case. Let us say you take the double dice throw experiment. And this probability of A is simply equal to cardinality of A divided by 36.

So, every element in the sample space is of the form A comma B, where A and B are between 1 to 6. So, probably the A cardinality of A by 36. Now I can ask the question, what is the probability that the sum of the dice is equal to let us say 6? How would you go about doing that? Well, the answer to that is, you simply count the number of outcomes, elementary outcomes, that sum equal to 6 divided by 6.

How many ways can you get a summation of 6? You can get summation, either from 1 comma 5 or 2 comma 4 or 3 comma 3 or 4 comma 2 or 5 comma 1. So, that is five ways you can get a 6 out of get sum of 6 from the 36 elements in sample space, by 36. So, this is a, counting is a powerful trick or powerful tool in your kitty for measuring probabilities, at least, for this special

probability measure where the probability measure is simply proportional to the cardinality of the set.

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So, here is a slightly more complicated example. Let us say you have, let us say you are playing scrabble and you have six characters with let's say, B, A, N, A, N, A. Let us say, you pick three scrabble coins, three scrabble pieces or squares at random. What is the probability? Probability that three pieces are A, A and A. So, this is a kind of typical question that you might see in probability that is, you have six scrabble coins corresponding to B, A and A and A and what is the probability of some event?

And just by symmetry, pretty much all events are going to be equally likely. All individual outcomes are equally likely. But let us make that clear. So, what is the sample space here? Well, for simplicity, let us say what you do is, you randomly permute, the 6 randomly order these 6 coins, and then you pick the first three, so that is the experiment, that is how you are making the thing more formal. In that case, your ω, your sample space is simply the set of all possible permutations.

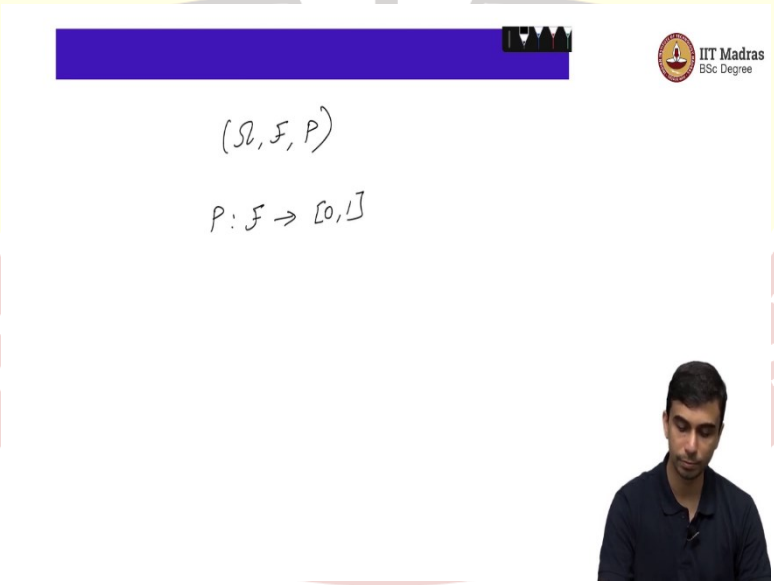
So, it is a 6 in fact that our cardinality of ω is going to be equal to 6 factorial. So, you can permute we can arrange the six coins in any order that you want that is there are 6 factorial ways in which you can arrange. That is your, and what you are going to do is you are going to pick the first three, in the, let us say, from the left to right, you are going to pick the first three pieces that and you want to know what is the probability that it is equal to A, A, A.

How many such events are there first? Number of what is a number of favorable outcomes? Number of outcomes, which can lead to the first three being A, A, A? So, first of all, you can immediately see that the ordering between the three A's do not matter. So, that is three factorial ways in which you can arrange the three As, and similarly, the ordering between the next 3 also do not matter.

So, there are three factorial and three factorial ways, where in which, out of the 6 factorial ways. If the 6 pieces were arranged that way, your first three pieces would be A, A, A. So, that is a number of favorable outcomes for calculating the probability of this particular event is 3 factor into 3 factorial. And total number of outcomes is 6 factorial, so, the probability of this event is simply equal to 3 factorial into 3 factorial by 6 factorial.

You can compute that using other ways also, but counting gives you a powerful tool for computing probabilities. At least in special cases of this probability measure the probability of A is simply cardinality of A by ω. Once again, note that every element here A is simply a subset of the sample space. So, this can never exceed 1, and the property of the null set is always going to be 0. So, this is always a valid probability measure.

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So, with that, we wrap up the basic probability space concept which an experiment is made is defined by a probability space which consists of three things, which is the sample space ω, a sigma algebra F, and the probability measure. Ω F is simply a collection of subsets of ω and P is simply a mapping from F to 0. So, with that, we wrap up the basic probability tools.